# Zach Sahlin

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## EDUCATION

**Gonzaga University** | Spokane, WA — August 2019 – May 2023
*BS in Computer Science, Minor in Applied Mathematics*
- GPA: 3.7, Magna Cum Laude

## WORK EXPERIENCE

**Software Engineer** | MAQ Software — *June 2024 – Present*
- Developed data pipelines using Databricks, Fabric, and Azure Data Factory, transforming user telemetry data of AI tools into curated datasets enabling data science and reporting for ROI measurements.
- Refactored and optimized spark ml pipeline in Azure Databricks to cut runtime by 95% and reduce monthly refresh delay from 3 days to overnight.
- Analyzed data in order to investigate and resolve data issues, drive requirements, and provide business insights.
- Spearheaded requirement gathering by collaborating with stakeholders and proactively raising issues and concerns in order to provide clear technical specs streamlining development.
- Managed multiple projects creating Power BI dashboards with up to 5000 monthly users for analyzing sales data for a cloud provider.
- Validated Spark notebooks and reports to verify project requirements were met and improve code stability.

**Teaching Assistant** | Gonzaga University — *August 2022 – May 2023*
- Assisted Dr. Yanping Zhang in Operating Systems, Computer Security, and Internet of Things courses
- Critically analyzed, tested, and graded programming and written assignments

## PROJECTS

**Stock Predictor** | Python, TensorFlow, AWS Lambda, AWS EventBridge, selenium — *Spring 2023*
- Designed a model to predict short term prices of stocks
- Created a data pipeline with web scrapers and APIs to collect historical and current data from various sources
- Leveraged AWS S3 to store and retrieve data
- Implemented a CI/CD pipeline with GitHub Actions to test and publish to AWS Lambda
- Developed a web app to display predictions updated daily

**Machine Learning for Real Estate** | Python, SQL, PyTorch, JS, Fast API — *Fall 2022 - Spring 2023*
- Collaborated with my Senior Design group to apply machine learning to real estate
- Collected, cleaned, and managed data on historical rental prices, crime data, and school scores
- Implemented and trained an LSTM to predict future rent prices
- Improved the REST API to decrease front end load time from 5 minutes to about 5 seconds
- Developed a web app to display historical prices and predictions in a graphical interface

**Resume Builder** | Python, JavaScript, TypeScript, React, Next.js, MongoDB, LaTeX, AWS Lambda — *Fall 2023*
- Created a React Next.js web app to allow users to input resume content and manage multiple resumes
- Developed an AWS Lambda function to format a resume in LaTeX and build the resume as a PDF
- Solved Lambda dependency and image size constraints by leveraging AWS ECR

## RELATED COURSEWORK

- Algorithms & Abstract Data Structures
- Machine Learning & Intelligent Systems
- Data Science Algorithms
- Operating Systems
- Database Management Systems
- Programming Languages & Compilers
- Natural Language Processing

## TECHNICAL SKILLS

- Languages: Python, SQL, Java, C++
- Tools: PySpark, Databricks, Fabric, Git, Linux, Docker, Jupyter, GitHub Actions, Postgres, AWS Lambda, AWS S3
- Libraries: TensorFlow, PyTorch, pandas, NumPy, scikit-learn, Matplotlib